

AERO70015/96012 Mathematics 3 Problem Set #1

Problem Sheet 1

1.1 PR1

Solve the Euler-Lagrange equation which makes the integral

$$I[y] = \int_0^{\pi/2} (2xyy' + y'^2) dx$$

stationary, given that $y = 0$ when $x = 0$ and $y = 1$ when $x = \pi/2$.

$$\frac{\partial f}{\partial y} - \frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) = 0$$

$$\therefore 2xy' - \frac{d}{dx} (2xy + 2y') = 2xy' - (2y + 2xy' + 2y'') = 0$$

$$\therefore y'' + y = 0$$

$$\text{Let } y = A \cos x + B \sin x$$

$$\text{when } x = 0, y = A = 0$$

$$\text{when } x = \frac{\pi}{2}, y = B = 1$$

$$\therefore y = \sin x$$

1.2 PR2

Find the minimising curves of the integral

$$I[\theta] = \int r^2 \left[1 + r^2 \left(\frac{d\theta}{dr} \right)^2 \right]^{1/2} dr. \quad \text{extremise } \int r^2 ds$$

$$f(r, \theta') \text{ where } \theta = u(r)$$

$$\therefore \frac{\partial f}{\partial \theta'} = \text{constant}$$

$$\therefore r^2 \theta' (1 + r^2 \theta'^2)^{-1/2} = \text{constant} = C$$

$$\therefore r^3 \theta'^2 = C^2 (1 + r^2 \theta'^2)$$

$$\therefore (r^3 - C^2 r^2) \theta'^2 = C^2$$

$$\therefore \theta'^2 = \frac{C^2}{r^3 - C^2 r^2}$$

$$\therefore \theta' = \frac{C}{r(r^2 - C^2)^{1/2}} \quad * C \text{ is arbitrary, } \pm \frac{C}{r(r^2 - C^2)^{1/2}} \text{ both fine}$$

$$\therefore \theta = C \int r^{-1} (r^2 - C^2)^{-1/2} dr$$

$$\text{Let } r^2 = C \sec t$$

$$\therefore 3r^2 dr = C \frac{0 - (-\sin t)}{\cos^3 t} dt = C \tan t \sec t dt$$

$$\therefore \theta = C \int \frac{1}{3} r^{-3} [(r^2)^2 - C^2]^{-1/2} C \tan t \sec t dt$$

$$= \frac{1}{3} \int \cos t (\sec^2 t - 1)^{-1/2} \tan t \sec t dt = \frac{1}{3} \int dt = \frac{1}{3} t + D$$

$$= \frac{1}{3} \cos^{-1} \left(\frac{C}{r^2} \right) + D$$

$$* \frac{1}{x \sqrt{x^2 + a^2}} \rightarrow \text{try } x = a \tan \theta$$

$$* \sqrt{x^2 + a^2} \rightarrow \text{try } x = a \tan \theta$$

$$\frac{1}{x \sqrt{x^2 - a^2}} \rightarrow \text{try } x = a \sec \theta$$

$$\sqrt{x^2 - a^2} \rightarrow \text{try } x = a \sec \theta$$

$$\frac{a}{x \sqrt{x^2 - a^2}} \rightarrow \text{try } x = a \cot \theta$$

$$\sqrt{a^2 - x^2} \rightarrow \text{try } x = a \cot \theta$$

1.3 PR3

Show that the extremal curves of the integral

$$I = \int \left[1 + \left(\frac{d\phi}{d\theta} \right)^2 \sin^2 \theta \right]^{1/2} d\theta$$

can be written in the form $\sin(\alpha - \phi) = \beta \cot \theta$, where α and β are constants.

spherical with $r=1$

$$x = \sin \phi \cos \theta$$

$$y = \sin \phi \sin \theta$$

$$z = \cos \phi$$

$$ds = \sqrt{dx^2 + dy^2 + dz^2}$$

$$= \sqrt{1 + \left(\frac{d\phi}{d\theta} \right)^2 \sin^2 \theta} d\theta$$

$$f(\theta, \phi') \text{ for } \phi = u(\theta)$$

$$\therefore \frac{\partial f}{\partial \phi'} = \text{constant}$$

$$\therefore \phi' \sin^2 \theta (1 + \phi'^2 \sin^2 \theta)^{-1/2} = A$$

$$\therefore \phi'^2 \sin^4 \theta = A^2 (1 + \phi'^2 \sin^2 \theta)$$

$$\therefore \phi' = \frac{A}{\sin \theta \sqrt{\sin^2 \theta - A^2}} \quad \therefore \phi = \int \frac{A}{\sin \theta \sqrt{\sin^2 \theta - A^2}} d\theta$$

$$\text{Let } u = \cot \theta = \frac{\cos \theta}{\sin \theta} = \frac{\sqrt{1 - \sin^2 \theta}}{\sin \theta} \quad \therefore \sin \theta = \frac{1}{\sqrt{u^2 + 1}}$$

$$\therefore du = -\operatorname{cosec}^2 \theta d\theta \quad \therefore d\theta = -\sin^2 \theta du$$

$$\therefore \phi = \int \frac{-A \sin^2 \theta}{\sin \theta \sqrt{\frac{1}{u^2 + 1} - A^2}} du$$

$$= \int \frac{-A}{\sqrt{u^2 + 1} \sqrt{\frac{1}{u^2 + 1} - A^2}} du$$

$$= \int \frac{-A}{\sqrt{1 - A^2(u^2 + 1)}} du = \int \frac{-A}{\sqrt{(1 - A^2) - A^2 u^2}} du = - \int \frac{1}{\sqrt{\frac{1 - A^2}{A^2} - u^2}} du$$

$$= -\sin^{-1} \left(u \sqrt{\frac{A^2}{1 - A^2}} \right) + \alpha$$

$$\therefore \sin(\alpha - \phi) = u \sqrt{\frac{A^2}{1 - A^2}} = \cot \phi \sqrt{\frac{A^2}{1 - A^2}} = \beta \cot \phi \text{ for } \beta = \sqrt{\frac{A^2}{1 - A^2}}$$

1.4 PR4

Find the stationary value of the integral

$$I = \int_0^1 y'^2 dx$$

subject to the constraint

$$J = \int_0^1 y dx = 1$$

and the end conditions $y(0) = 0$ and $y(1) = 0$.

$$f + \lambda g = y'^2 + \lambda y = f(y, y')$$

$$\therefore (f + \lambda g) - y' \frac{\partial}{\partial y'} (f + \lambda g) = \text{constant}$$

$$\therefore y'^2 + \lambda y - y'(2y') = \lambda y - y'^2 = C$$

$$\therefore \frac{dy}{dx} = (\lambda y - C)^{\frac{1}{2}}$$

$$\therefore \int (\lambda y - C)^{-\frac{1}{2}} dy = \int dx$$

$$\therefore \frac{2}{\lambda} (\lambda y - C)^{\frac{1}{2}} = x + D$$

$$\therefore \lambda y - C = \frac{\lambda^2}{4} (x + D)^2$$

$$\therefore y = \frac{\lambda}{4} (x + D)^2 + \frac{C}{\lambda}$$

$$\text{for } y(0) = 0, \quad 0 = \frac{\lambda}{4} D^2 + \frac{C}{\lambda}$$

$$\text{for } y(1) = 0, \quad 0 = \frac{\lambda}{4} (1 + D)^2 + \frac{C}{\lambda}$$

$$\therefore D = -\frac{1}{2}$$

$$\therefore C = -\frac{\lambda^2}{16}$$

$$\therefore y = \frac{\lambda}{4} \left(x - \frac{1}{2}\right)^2 - \frac{\lambda}{16} = \frac{\lambda}{4} \left(x^2 - x + \frac{1}{4}\right) - \frac{\lambda}{16}$$

$$\therefore \int_0^1 y \, dx = 1$$

$$\therefore \left[\frac{\lambda}{4} \left(\frac{1}{3}x^3 - \frac{1}{2}x^2 + \frac{1}{4}x\right) - \frac{\lambda}{16}x \right]_0^1 = -\frac{\lambda}{24} = 1$$

$$\therefore \lambda = -24$$

$$\therefore y = -6 \left(x - \frac{1}{2}\right)^2 + \frac{3}{2} = -6x^2 + 6x$$

$$\therefore y' = -12x + 6$$

$$\therefore y'^2 = 144 \left(x - \frac{1}{2}\right)^2 = 144x^2 - 144x + 36$$

$$\therefore I = \int_0^1 y'^2 \, dx = \left[48x^3 - 72x^2 + 36x \right]_0^1 = 12$$

1.5 PR5

Prove that the extremal curve $y = y(x)$ of the integral

$$I = \int_1^2 \left[x^2 \left(\frac{dy}{dx} \right)^2 + 2y^2 \right] dx$$

which passes through the points (1, 0) and (2, 1) in the $x - y$ plane is given by $y = \frac{4}{7} \left(x - \frac{1}{x^2} \right)$.

If the constraint $\int_1^2 \frac{y}{x} dx = \frac{1}{4}$ is added to the problem, find the new extremal curve of I .

$$\frac{\partial \mathcal{L}}{\partial y} - \frac{d}{dx} \left(\frac{\partial \mathcal{L}}{\partial y'} \right) = 0$$

$$\therefore 4y - \frac{d}{dx} (2x^2 y') = 4y - 4xy' - 2x^2 y'' = 0$$

$$\therefore x^2 y'' + 2xy' - 2y = 0$$

$$\text{let } y = A \left(x - \frac{1}{x^2} \right) \quad \therefore 2y = A \left(2x - 2\frac{1}{x^2} \right)$$

$$\therefore y' = A \left(1 + 2\frac{1}{x^3} \right) \quad \therefore 2xy' = A \left(2x + 4\frac{1}{x^2} \right)$$

$$\therefore y'' = A \left(-6\frac{1}{x^3} \right) \quad \therefore x^2 y'' = A \left(-6\frac{1}{x} \right)$$

At (1, 0) and (2, 1)

$$\begin{cases} 0 = A(1 - 1) = 0 \\ 1 = A \left(2 - \frac{1}{4} \right) = \frac{7}{4}A \end{cases}$$

$$\therefore A = \frac{4}{7}$$

$$\therefore y = \frac{4}{7} \left(x - \frac{1}{x^2} \right)$$

$$* \text{ let } y = x^n$$

$$\therefore n(n-1)x^n + 2nx^n - 2x^n = 0$$

$$\therefore n^2 + n - 2 = 0$$

$$\therefore (n-1)(n+2) = 0$$

$$\therefore n = 1 \text{ and } -2$$

$$\therefore y = Ax + Bx^{-2}$$

if constraint being added, $f + \lambda g = x^2 y'^2 + 2y^2 + \lambda \frac{y}{x}$

$$\therefore (4y + \lambda \frac{1}{x}) - \frac{d}{dx} (2x^2 y') = 4y + \lambda \frac{1}{x} - 4xy' - 2x^2 y'' = 0$$

$$\therefore x^2 y'' + 2xy' - 2y = \frac{\lambda}{2} \frac{1}{x}$$

$$\text{let } y = B(x - \frac{1}{x^2} + \frac{1}{x})$$

$$\therefore y' = B(1 + 2\frac{1}{x^3} - \frac{1}{x^2}) \quad \therefore y'' = B(-6\frac{1}{x^4} + 2\frac{1}{x^3})$$

$$\therefore -2B\frac{1}{x} = \frac{\lambda}{2} \frac{1}{x}$$

$$\therefore B = -\frac{\lambda}{4}$$

$$\therefore y = -\frac{\lambda}{4}(x - \frac{1}{x^2} + \frac{1}{x})$$

$$\int_1^2 \frac{y}{x} dx = \int_1^2 -\frac{\lambda}{4}(1 - \frac{1}{x^3} + \frac{1}{x}) dx$$

$$= -\frac{\lambda}{4} [x + \frac{1}{2x^2} - \frac{1}{3x^3}]_1^2$$

$$= -\frac{\lambda}{4} (\frac{25}{12} - \frac{7}{6}) = -\frac{11}{48}\lambda = \frac{1}{4}$$

$$\therefore \lambda = -\frac{12}{11}$$

$$\therefore y = \frac{3}{11}(x + \frac{1}{x} - \frac{1}{x^2})$$

$$\text{let } y = Ax + B\frac{1}{x} + C\frac{1}{x^2}$$

$$\therefore y' = A - B\frac{1}{x^2} - 2C\frac{1}{x^3}$$

$$\therefore y'' = 2B \frac{1}{x^3} + 6C \frac{1}{x^4}$$

$$\begin{aligned} \therefore (2B \frac{1}{x} + 6C \frac{1}{x^2}) + (2Ax - 2B \frac{1}{x} - 4C \frac{1}{x^2}) + (-2Ax - 2B \frac{1}{x} - 2C \frac{1}{x^2}) \\ = -2B \frac{1}{x} = \frac{\lambda}{2} \frac{1}{x} \end{aligned}$$

$$\therefore B = -\frac{\lambda}{4}$$

At (1,0), (2,1)

$$\begin{cases} 0 = A + B + C & \therefore C = -(A+B) \\ 1 = 2A + \frac{1}{2}B + \frac{1}{4}C = \frac{7}{4}A + \frac{1}{4}B & \therefore A = \frac{4}{7} - \frac{1}{7}B \end{cases}$$

$$\therefore y = (\frac{4}{7} - \frac{1}{7}B)x + B \frac{1}{x} - (\frac{4}{7} + \frac{6}{7}B) \frac{1}{x^2}$$

$$\begin{aligned} \therefore \int_1^2 \frac{y}{x} dx &= \int_1^2 (\frac{4}{7} - \frac{1}{7}B) + B \frac{1}{x^2} - (\frac{4}{7} + \frac{6}{7}B) \frac{1}{x^3} dx \\ &= \left[(\frac{4}{7} - \frac{1}{7}B)x - B \frac{1}{x} + (\frac{2}{7} + \frac{3}{7}B) \frac{1}{x^2} \right]_1^2 \\ &= (\frac{4}{7} - \frac{1}{7}B) + \frac{1}{2}B + (-\frac{3}{14} - \frac{9}{28}B) \\ &= \frac{1}{28}B + \frac{5}{14} = \frac{1}{4} \end{aligned}$$

$$\therefore B = -3$$

$$\therefore y = x - 3 \frac{1}{x} + 2 \frac{1}{x^2}$$

1.6 PR6

(a) Prove that the extremal curve $y = y(x)$ of the integral

$$I = \int_0^{2\pi} \left[m^2 y^2 - \left(\frac{dy}{dx} \right)^2 \right] dx$$

(where m is an integer) satisfying the conditions $y = 1$ for $x = 0$ and $dy/dx = \pi/2$ for $x = 2\pi$, and the constraint

$$\int_0^{2\pi} y(x) \cos(nx) dx = \frac{\pi}{2}, \quad \text{where } n \text{ is an integer,}$$

is given by

$$y = \frac{1}{2} \left(\cos(mx) + \frac{\pi}{m} \sin(mx) + \cos(nx) \right) \quad \text{provided } m \neq n.$$

(b) Find the extremal curve of I subject to the same conditions for the special case $m = n$. (Note that this is not just a matter of substituting $m = n$ in the expression from part a!)

$$(a) \quad f + \lambda g = m^2 y^2 - y'^2 + \lambda y \cos(nx)$$

$$\therefore \frac{\partial}{\partial y} (f + \lambda g) - \frac{d}{dx} \frac{\partial}{\partial y'} (f + \lambda g) = 0$$

$$\therefore 2m^2 y + \lambda \cos(nx) - \frac{d}{dx} (-2y') = 0$$

$$\therefore 2m^2 y + \lambda \cos(nx) + 2y'' = 0 \quad (1)$$

$$\text{Let } y = \frac{1}{2} [\cos(mx) + \frac{\pi}{m} \sin(mx) + \cos(nx)]$$

$$\therefore y' = \frac{1}{2} [-m \sin(mx) + \pi \cos(mx) - n \sin(nx)]$$

$$\therefore y'' = \frac{1}{2} [-m^2 \cos(mx) - m\pi \sin(mx) - n^2 \cos(nx)]$$

$$\text{for } x = 0, \quad y = \frac{1}{2} (1 + 1) = 1$$

$$\text{for } x = 2\pi, \quad y' = \frac{1}{2} \pi$$

$\therefore$ substitute into equation ①,

$$\cos(mx) (m^2 - m^2) + \sin(mx) (m\pi - m\pi)$$

$$+ \cos(nx) (\lambda + m^2 - n^2) = 0$$

$$\therefore \lambda = n^2 - m^2$$

$$\int_0^{2\pi} y \cos(nx) dx$$

$$= \int_0^{2\pi} \frac{1}{2} \left[\cos(mx) \cos(nx) + \frac{\pi}{m} \sin(mx) \cos(nx) + \cos^2(nx) \right] dx$$

$$= \frac{1}{4} \int_0^{2\pi} \cos[(m-n)x] + \cos[(m+n)x] dx$$

$$+ \frac{\pi}{4m} \int_0^{2\pi} \sin[(m-n)x] + \sin[(m+n)x] dx$$

$$+ \frac{1}{4} \int_0^{2\pi} 1 + \cos[(m+n)x] dx$$

$$= 0 + 0 + \frac{1}{4} \cdot 2\pi = \frac{1}{2}\pi$$

(b) for $m = n$,

$$2y'' + 2m^2y = -\lambda \cos(mx)$$

$$\text{let } y = A \sin(mx) + B \cos(mx) + Cx \sin(mx)$$

$$\therefore 2y'' + 2m^2y = 4Cm \cos(mx) = -\lambda \cos(mx)$$

$$\therefore C = -\frac{\lambda}{4m}$$

$$\text{for } x=0, \quad y = B = 1$$

$$\text{for } x=2\pi, \quad y' = mA + 2\pi mC = \frac{\pi}{2}$$

$$\text{for } \int_0^{2\pi} A \sin(mx) \cos(mx) + \cos^2(mx) + Cx \sin(mx) \cos(mx) \, dx$$

$$= \int_0^{2\pi} \frac{1}{2} A \sin(2mx) + \frac{1}{2} [1 + \cos(2mx)] + \frac{1}{2} Cx \sin(2mx) \, dx$$

$$= \frac{1}{2} A \left[-\frac{1}{2m} \cos(2mx) \right]_0^{2\pi} + \frac{1}{2} \left[x + \frac{1}{2m} \sin(2mx) \right]_0^{2\pi}$$

$$+ \frac{1}{2} C \left[-\frac{x}{2m} \cos(2mx) + \frac{1}{4m^2} \sin(2mx) \right]_0^{2\pi}$$

$$= \pi + \frac{1}{2} C \left(-\frac{\pi}{m} \right) = \frac{1}{2} \pi$$

$$\therefore \pi = \frac{\pi}{m} C \quad \therefore C = m$$

$$\therefore A = \frac{\pi}{2m} - 2\pi C = \frac{\pi}{2m} - 2\pi m$$

$$\therefore y = \left(\frac{\pi}{2m} - 2\pi m \right) \sin(mx) + \cos(mx) + mx \sin(mx)$$

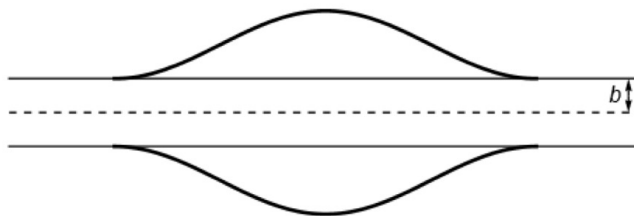
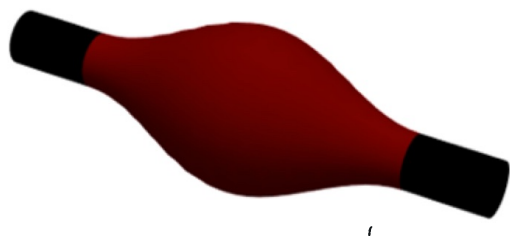
1.7 PR7

The shape of an oil drop on a fibre minimises the surface area for a given volume, subject to the constraint that where the drop contacts the fibre, the surface of the oil and the surface of the fibre are parallel. If the radius of the fibre is b , and the oil drop is the volume of revolution of the curve $z = z(x)$, then the total volume of the drop is

$$V = \pi \int (z^2 - b^2) dx, \text{ and the total surface area is } A = 2\pi \int z (1 + z'^2)^{1/2} dx.$$

The function $z(x)$ is the extremal curve of $G = A - \lambda V$.

Find, in terms of b and a Lagrange multiplier λ , a first order ODE satisfied by z .



$$V = \int A \, dx = \int \pi r^2 \, dx = \pi \int (z^2 - b^2) \, dx = V_0$$

$$A = \int 2\pi r \, ds = 2\pi \int z \sqrt{1 + z'^2} \, dx$$

$$G = A - \lambda V$$

$$\therefore g = 2z(1 + z'^2)^{\frac{1}{2}} - \lambda(z^2 - b^2)$$

To minimise A for given V_0 , $g - z' \frac{\partial g}{\partial z'} = \text{constant}$

$$\therefore 2z(1 + z'^2)^{\frac{1}{2}} - \lambda(z^2 - b^2) - z' [2zz'(1 + z'^2)^{-\frac{1}{2}}]$$

$$= 2z(1 + z'^2)^{-\frac{1}{2}} - \lambda(z^2 - b^2) = k$$

$$\text{when } z = b, \quad z' = 0$$

$$\therefore 2b = k$$

$$\therefore 2z(1 + z'^2)^{-\frac{1}{2}} - \lambda(z^2 - b^2) = 2b$$

$$\therefore 1 + z'^2 = \frac{4z^2}{[\lambda(z^2 - b^2) + 2b]^2}$$

$$\therefore z' = \left\{ \frac{4z^2}{[\lambda(z^2 - b^2) + 2b]^2} - 1 \right\}^{\frac{1}{2}}$$

1.8 Past Exam Question

(a) By setting up and solving the Euler-Lagrange equation, find the extremiser $y(x)$ of the integral

$$I = \int_0^{\ln 2} (y'^2 - y y' + y^2) dx,$$

where $y' = dy/dx$, subject to the end conditions $y(0) = y(\ln 2) = 3$.

(b) The constraint

$$\int_0^{\ln 2} x y' dx = -2$$

is now applied. Show that the new extremiser of I , subject to this constraint, can be written

$$y = (1 - \lambda)(e^x + 2e^{-x}) + 3\lambda$$

and find an expression for λ in terms of $\ln 2$.

$$(a) \quad f - y' \frac{\partial f}{\partial y'} = \text{constant}$$

$$\therefore y'^2 - y y' + y^2 - y'(2y' - y)$$

$$= -y'^2 + y^2 = k = \text{constant}$$

$$\text{Let } y = Ae^x + Be^{-x}$$

$$\therefore y' = Ae^x - Be^{-x}$$

$$\therefore -y'^2 + y^2 = -(A^2 e^{2x} - 2AB + B^2 e^{-2x}) + (A^2 e^{2x} + 2AB + B^2 e^{-2x})$$

$$= 4AB = k$$

$$\text{for } y(0) = 3, \quad 3 = A + B \quad \therefore 6 = 2A + 2B$$

$$\text{for } y(\ln 2) = 3, \quad 3 = 2A + \frac{1}{2}B$$

$$\therefore B = 2 \quad \therefore A = 1$$

$$\therefore k = 4AB = 8 \quad \therefore y = e^x + 2e^{-x}$$

$$b) f + \alpha g = y'^2 - yy' + y^2 + \alpha(xy')$$

To find extremise,

$$-y' + 2y - \frac{d}{dx}(2y' - y + \alpha x) = 0$$

$$\therefore -y' + 2y - 2y'' + y' - \alpha = 0 \quad \therefore -2y'' + 2y = \alpha$$

$$\text{let } y = Ae^x + Be^{-x} + C$$

$$\therefore y'' = Ae^x + Be^{-x}$$

$$\therefore -2y'' + 2y = 2C = \alpha \quad \therefore C = \frac{1}{2}\alpha$$

$$\text{for } y(0) = 3, A + B + \frac{1}{2}\alpha = 3 \quad \therefore 2A + 2B + \alpha = 6$$

$$\text{for } y(\ln 2) = 3, 2A + \frac{1}{2}B + \frac{1}{2}\alpha = 3$$

$$\therefore \frac{3}{2}B + \frac{1}{2}\alpha = 3 \quad \therefore B = 2 - \frac{1}{3}\alpha \quad \therefore A = 1 - \frac{1}{6}\alpha$$

$$\therefore y = (1 - \frac{1}{6}\alpha)e^x + (2 - \frac{1}{3}\alpha)e^{-x} + \frac{1}{2}\alpha$$

let $\lambda = \frac{1}{6}\alpha$, therefore,

$$y = (1 - \lambda)(e^x + 2e^{-x}) + 3\lambda \quad \therefore y' = (1 - \lambda)(e^x - 2e^{-x})$$

$$\begin{aligned} \int_0^{\ln 2} xy' dx &= (1 - \lambda) \int_0^{\ln 2} xe^x - 2xe^{-x} dx \\ &= (1 - \lambda) [xe^x - e^x + 2xe^{-x} + 2e^{-x}]_0^{\ln 2} \\ &= (1 - \lambda)(3\ln 2 - 2) = -2 \end{aligned}$$

$$\therefore \lambda = \frac{2}{3\ln 2 - 2} + 1 \quad \therefore \lambda = \frac{3\ln 2}{3\ln 2 - 2}$$